

DSA Practical No: 02

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Div: SY3

Class: C3

INPUT :

```
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int key)
{
    int mid;

    if (low > high)
        return -1;

    mid = (low + high) / 2;

    if (arr[mid] == key)
        return mid;

    if (key < arr[mid])
        return binarySearch(arr, low, mid - 1, key);

    return binarySearch(arr, mid + 1, high, key);
}

int main()
{
    int arr[100];

    int n, i, key, result;
```

```
char ch;

do
{
    printf(" RECURSIVE BINARY SEARCH\n");

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d sorted elements:\n", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);

    result = binarySearch(arr, 0, n - 1, key);

    if (result == -1)
    {
        printf("Element %d is not found in the array.\n", key);
    }
    else
    {

```

```

        printf("Element %d is found at index %d.\n", key, result);
    }

    printf("\nDo you want to continue? (y/n): ");

    scanf(" %c", &ch);

} while (ch == 'y' || ch == 'Y');

printf("\nProgram ended.\n");

return 0;
}

```

OUTPUT :

```

PS D:\dsa\Experiment-02-Recursive-Binary-Search> .\program.exe
    RECURSIVE BINARY SEARCH
Enter number of elements: 5
Enter 5 sorted elements:
1
2
3
4
5
Enter element to search: 2
Element 2 is found at index 1.

Do you want to continue? (y/n): y
    RECURSIVE BINARY SEARCH
Enter number of elements: 5
Enter 5 sorted elements:
5
4
3
2
1
Enter element to search: 2
Element 2 is not found in the array.

Do you want to continue? (y/n): 

```